The Æther-flow Interpretation of Relativity

Constraint-Assisted U-Sector Quadratic Elimination and Generalized Exact-Closure Conditions

Abstract

This manuscript carries the constraint-assisted longitudinal U-sector family through its next fixed derivational task: full flat-background quadratic elimination and extraction of the generalized exact-closure conditions. The scope remains disciplined. The note does not claim a completed substrate derivation of Einsteinian gravity. It computes only the quadratic status of the candidate family already selected in the active research plan.

The new auxiliary scalar C changes the elimination problem in one precise way. After the coupled (χ, C) block is integrated out, the scalar source combination $\alpha X + \beta_I q^I$, with $X \equiv \partial_0\sigma - \sigma_0\phi$, feeds back into the Q-sector as a rank-one deformation of the operator $k^2\delta_{IJ} + \widehat{M}_{IJ}$. Defining

$$\Delta_C \equiv 1 - \kappa_\theta M_C^2, \qquad \mathcal{D}_C(k^2) = \Delta_C + \kappa_\theta\, \beta_I \big(k^2\delta + \widehat{M}\big)^{-1\,IJ} \beta_J,$$

the reduced scalar bridge channel is governed by three momentum-dependent coefficients $\mathcal{M}_C(k^2)$, $\Gamma_C(k^2)$, and $\mathcal{B}_C(k^2)$, and the observer-accessible scalar bridge self-energy becomes

$$\Pi_\phi^{(C)}(\omega, \mathbf{k}) = \sigma_0^2 \frac{\mathcal{N}_C(k^2)}{\mathcal{M}_C(k^2)\omega^2 + \mathcal{B}_C(k^2)}, \qquad \mathcal{N}_C = \mathcal{M}_C\mathcal{B}_C - \Gamma_C^2.$$

Hence the flat-background exact-closure burden is no longer carried by $\mathcal{B}(k^2)$ alone. On the generic nonresonant branch, it is carried by the generalized numerator $\mathcal{N}_C(k^2)$. Its infrared coefficients $\mathcal{N}_{C,0}$ and $\mathcal{N}_{C,2}$ give the generalized exact-closure conditions derived here. The separate question of whether those conditions admit a healthy nondegenerate solution is left to the next manuscript.

1. Introduction

The active exact-closure line of Æther-flow Interpretation of Relativity is already fixed by the foundations, dynamics, consistency, relativistic-recovery, flow-geometry, and substrate-bridge manuscripts. On the derivational side, the project has now passed through three sharper filters. First, the minimal candidate action was written explicitly. Second, the coefficient-level linearized-consistency note reduced its flat-background exact-closure burden to the scalar operator $\mathcal{B}(k^2)$. Third, the minimal stable-sign no-go note and the derivative-mixing branch consistency note showed that neither the minimal branch nor the first nonminimal S/Q derivative-mixing branch yields the sought nondegenerate flat-background exact-closure solution.

That sequence fixed the next target. The branch-selection note for the constraint-assisted longitudinal U-sector family showed why the previous no-go decompositions no longer apply once the divergence of the ordered-flow field is sourced by the order and Q sectors through an auxiliary

scalar C. What that note deliberately did not do was the full Phase 3 elimination. The present manuscript performs that elimination.

The result is the exact analogue of the earlier coefficient-level reductions, but with one decisive change. The exact-closure burden no longer collapses directly to the old scalar operator $\mathcal{B}(k^2)$ or to the derivative-mixing generalization $\mathcal{B}_\rho(k^2)$. Instead it is carried by a new generalized numerator built from the coupled longitudinal U-sector and Q-sector response.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays inside that boundary. It does not promote the constraint-assisted branch to established theory. It determines the branch's full quadratic elimination and the corresponding generalized exact-closure conditions.

2. Constraint-Assisted Branch and Background Variables

The constraint-assisted longitudinal U-sector family modifies the auxiliary sector by

$$\Delta\mathcal{L}_C = C\Big(\nabla_A U^A - \alpha\big(U^B\partial_B S - \sigma_0\big) - \beta_I Q^I\Big) - \frac{M_C^2}{2}C^2. \tag{1}$$

The full branch auxiliary sector is therefore

$$\mathcal{L}_{\rm aux}^{(C)} = \mathcal{L}_{\rm aux} + \Delta\mathcal{L}_C, \tag{2}$$

where $\mathcal{L}_{\rm aux}$ is the minimal auxiliary sector already defined in the substrate-bridge and coefficient-level notes.

As in the previous flat-background analyses, use the admissible homogeneous bridge background

$$\bar{G}_{AB} = \delta_{AB}, \qquad \bar{U}^A = \delta^A{}_0, \qquad \bar{S} = \sigma_0 X^0, \qquad \bar{Q}^I = 0, \tag{3}$$

for which

$$\bar{\mathfrak{g}}_{AB} = \eta_{AB} = \mathrm{diag}(-1,1,1,1). \tag{4}$$

Write perturbations as

$$G_{AB} = \delta_{AB} + H_{AB}, \qquad U^A = \bar{U}^A + V^A, \qquad S = \sigma_0 X^0 + \sigma, \qquad Q^I = q^I. \tag{5}$$

The unit constraint again gives

$$V^0 = -\frac{1}{2}H_{00}, \tag{6}$$

and the bridge perturbation is

$$h_{00} = -H_{00}, \qquad h_{0i} = -H_{0i} - 2V_i, \qquad h_{ij} = H_{ij}. \tag{7}$$

Decompose

$$V_i = V_i^T + \partial_i \chi, \qquad \partial_i V_T^i = 0, \tag{8}$$

and define

$$X \equiv \partial_0 \sigma - \sigma_0 \phi, \qquad \Theta \equiv \nabla^2 \chi + \frac{1}{2}\partial_0 H, \tag{9}$$

with $h_{00} = -2\phi$ and $H = \delta^{ij} h_{ij}$.

3. Full Quadratic Auxiliary Lagrangian

Proposition 1 (Quadratic auxiliary coefficients of the constraint-assisted branch)**.** *On the flat background* (3)–(4)*, the full quadratic auxiliary Lagrangian is*

$$\begin{aligned}\mathcal{L}^{(2)}_{\mathrm{aux},C} = & -\frac{m_S^2}{2}X^2 - \frac{\kappa_S}{2}\,\partial_i\sigma\,\partial_i\sigma - \frac{1}{2}\mu_S^2\sigma^2 - \mu_{SI}\sigma q^I \\ & - \frac{1}{2}q^I\Big(-\nabla^2\delta_{IJ} + \widehat{M}_{IJ}\Big)q^J - \frac{\kappa_\theta}{2}\Theta^2 + C\big(\Theta - \alpha X - \beta_I q^I\big) - \frac{M_C^2}{2}C^2 \\ & - \frac{\kappa_\Omega}{4}\,\partial_{[i}\Big(S_{j]} + V_{j]}^T\Big)\,\partial^{[i}\Big(S^{j]} + V_T^{j]}\Big)\,. \end{aligned} \tag{10}$$

Proof. The coefficient-level linearized-consistency note already derived the minimal quadratic auxiliary sector, and the branch-selection note already derived the quadratic expansion of the new interaction (1). At this order the only additional term is

$$\Delta\mathcal{L}^{(2)}_C = C\big(\Theta - \alpha X - \beta_I q^I\big) - \frac{M_C^2}{2}C^2.$$

Adding that contribution to the previously derived minimal quadratic coefficients yields (10). □

Remark 1. *The tensor bridge sector remains the ordinary Fierz–Pauli sector of the bridge metric exactly as in the previous notes. All new quadratic structure is carried by the scalar longitudinal U-sector and its feedback into the Q-sector.*

4. Exact Elimination of the Sourced Longitudinal U-Sector

Definition 1 (Generic nonresonant constraint-assisted branch)**.** *For the present quadratic elimination, the generic nonresonant branch is the part of parameter space satisfying*

$$\Delta_C \equiv 1 - \kappa_\theta M_C^2 \neq 0. \tag{11}$$

The locus $\Delta_C = 0$ is a separate degenerate branch and is not part of the generic elimination derived below.

Proposition 2 (On-shell elimination of the (χ, C) block)**.** *On the decaying flat branch with no*

harmonic zero mode and $\Delta_C \neq 0$, the coupled (χ, C) equations imply

$$\Theta = \frac{\alpha X + \beta_I q^I}{\Delta_C}, \qquad C = \frac{\kappa_\theta(\alpha X + \beta_I q^I)}{\Delta_C}. \tag{12}$$

Substituting back into (10) *gives the on-shell longitudinal contribution*

$$\mathcal{L}^{(2)}_{\Theta C,\text{on-shell}} = -\frac{\kappa_\theta}{2\Delta_C}\left(\alpha X + \beta_I q^I\right)^2. \tag{13}$$

Proof. Varying (10) with respect to C gives

$$\Theta - \alpha X - \beta_I q^I - M_C^2 C = 0. \tag{14}$$

Varying with respect to χ gives

$$\nabla^2\big(-\kappa_\theta\Theta + C\big) = 0. \tag{15}$$

On the decaying branch, (15) implies $C = \kappa_\theta\Theta$. Substituting into (14) yields

$$\Delta_C\,\Theta = \alpha X + \beta_I q^I,$$

which gives (12). Substituting (12) back into the (Θ, C) part of (10) gives (13). □

Corollary 1 (Unchanged transverse U-sector elimination)**.** *On the same decaying branch, the transverse equation is unchanged:*

$$V_T^i = -S^i. \tag{16}$$

Consequently the κ_Ω sector contributes no residual flat-background bridge self-energy at quadratic order.

Proof. The new C-interaction contains no V_T^i. Therefore the transverse U-sector variation is exactly the same as in the minimal and derivative-mixing branches. □

After eliminating (χ, C), the scalar/Q part of (10) becomes

$$\begin{aligned}\mathcal{L}^{(2)}_{\sigma q,C} = &-\frac{1}{2}\left(m_S^2 + \frac{\alpha^2\kappa_\theta}{\Delta_C}\right)X^2 - \frac{1}{2}\sigma\Big(\kappa_S(-\nabla^2) + \mu_S^2\Big)\sigma - \mu_{SI}\sigma q^I \\ &-\frac{1}{2}q^I\left[\mathcal{A}_{IJ}(-\nabla^2) + \frac{\kappa_\theta}{\Delta_C}\beta_I\beta_J\right]q^J - \frac{\alpha\kappa_\theta}{\Delta_C}\,X\,\beta_I q^I,\end{aligned} \tag{17}$$

where

$$\mathcal{A}_{IJ}(-\nabla^2) \equiv (-\nabla^2)\delta_{IJ} + \widehat{M}_{IJ}. \tag{18}$$

5. Elimination of the Q-Sector and Reduced Scalar Action

Definition 2 (Constraint-assisted response operators)**.** *Define the scalar operators*

$$\mathcal{G}(-\nabla^2) \equiv \beta_I \mathcal{A}^{-1\,IJ}(-\nabla^2)\beta_J, \tag{19}$$

$$\mathcal{Z}(-\nabla^2) \equiv \mu_{SI} \mathcal{A}^{-1\,IJ}(-\nabla^2)\beta_J, \tag{20}$$

$$\Sigma_q(-\nabla^2) \equiv \mu_{SI} \mathcal{A}^{-1\,IJ}(-\nabla^2)\mu_{SJ}, \tag{21}$$

$$\mathcal{B}(-\nabla^2) \equiv \kappa_S(-\nabla^2) + \mu_S^2 - \Sigma_q(-\nabla^2), \tag{22}$$

$$\mathcal{D}_C(-\nabla^2) \equiv \Delta_C + \kappa_\theta \mathcal{G}(-\nabla^2). \tag{23}$$

Lemma 1 (Rank-one inverse of the deformed Q-operator)**.** *Let*

$$\mathcal{A}^{(C)}_{IJ}(-\nabla^2) \equiv \mathcal{A}_{IJ}(-\nabla^2) + \frac{\kappa_\theta}{\Delta_C}\beta_I\beta_J. \tag{24}$$

On the generic nonresonant branch, its inverse is

$$\left(\mathcal{A}^{(C)}\right)^{-1\,IJ} = \mathcal{A}^{-1\,IJ} - \frac{\kappa_\theta}{\mathcal{D}_C}\,\mathcal{A}^{-1\,IK}\beta_K\beta_L\mathcal{A}^{-1\,LJ}. \tag{25}$$

Proof. Equation (24) is a rank-one deformation of $\mathcal{A}_{IJ}$. The Sherman–Morrison identity gives

$$\left(\mathcal{A}^{(C)}\right)^{-1} = \mathcal{A}^{-1} - \frac{\frac{\kappa_\theta}{\Delta_C}\,\mathcal{A}^{-1}\beta\beta^\top\mathcal{A}^{-1}}{1 + \frac{\kappa_\theta}{\Delta_C}\,\beta^\top\mathcal{A}^{-1}\beta}.$$

Using $\beta^\top\mathcal{A}^{-1}\beta = \mathcal{G}$ and $\mathcal{D}_C = \Delta_C + \kappa_\theta\mathcal{G}$ yields (25). □

Proposition 3 (Exact Q-sector elimination)**.** *The q^I Euler–Lagrange equation derived from* (17) *is*

$$\mathcal{A}^{(C)}_{IJ}(-\nabla^2)\,q^J = -\mu_{SI}\sigma - \frac{\alpha\kappa_\theta}{\Delta_C}\,\beta_I X. \tag{26}$$

Hence

$$q^I = -\mathcal{A}^{-1\,IJ}\mu_{SJ}\sigma - \frac{\kappa_\theta}{\mathcal{D}_C}\,\mathcal{A}^{-1\,IJ}\beta_J(\alpha X - \mathcal{Z}\,\sigma). \tag{27}$$

Proof. Equation (26) is immediate from (17). Applying (25) to the source on the right-hand side gives (27). □

Proposition 4 (Reduced scalar bridge channel)**.** *After eliminating q^I, the scalar bridge sector becomes*

$$\mathcal{L}^{(2)}_{\sigma,\mathrm{eff},C} = -\frac{1}{2}\,X\,\mathcal{M}_C(-\nabla^2)\,X + \sigma\,\Gamma_C(-\nabla^2)\,X - \frac{1}{2}\,\sigma\,\mathcal{B}_C(-\nabla^2)\,\sigma, \tag{28}$$

with

$$\mathcal{M}_C(-\nabla^2) = m_S^2 + \frac{\alpha^2\kappa_\theta}{\mathcal{D}_C(-\nabla^2)}, \tag{29}$$

$$\Gamma_C(-\nabla^2) = \frac{\alpha\kappa_\theta\,\mathcal{Z}(-\nabla^2)}{\mathcal{D}_C(-\nabla^2)}, \tag{30}$$

$$\mathcal{B}_C(-\nabla^2) = \mathcal{B}(-\nabla^2) + \frac{\kappa_\theta\,\mathcal{Z}(-\nabla^2)^2}{\mathcal{D}_C(-\nabla^2)}. \tag{31}$$

Proof. Substituting (27) into (17) and using (25) gives

$$\frac{1}{2}(\mu_{SI}\sigma + \tfrac{\alpha\kappa_\theta}{\Delta_C}\beta_I X)\,(\mathcal{A}^{(C)})^{-1\,IJ}(\mu_{SJ}\sigma + \tfrac{\alpha\kappa_\theta}{\Delta_C}\beta_J X)$$

for the on-shell Q-sector contribution. Evaluating the three resulting terms with (25) yields the coefficients (29)–(31), which gives (28). □

Corollary 2 (Minimal-branch reversion)**.** *If $\alpha = \beta_I = 0$, then*

$$\mathcal{M}_C = m_S^2, \qquad \Gamma_C = 0, \qquad \mathcal{B}_C = \mathcal{B}, \tag{32}$$

so the reduced scalar bridge channel contracts back to the minimal coefficient-level result.

6. Generalized Bridge Self-Energy and Exact-Closure Conditions

Proposition 5 (Generalized scalar bridge self-energy)**.** *Up to a boundary term, the mixed term in* (28) *satisfies*

$$\sigma\,\Gamma_C(-\nabla^2)\,X \simeq -\sigma_0\,\sigma\,\Gamma_C(-\nabla^2)\,\phi. \tag{33}$$

After integrating out σ, the observer-accessible scalar bridge correction is therefore

$$\Delta\mathcal{L}^{(2)}_{\phi,C}(\omega,\mathbf{k}) = -\frac{1}{2}\,\Pi^{(C)}_\phi(\omega,\mathbf{k})\,|\phi(\omega,\mathbf{k})|^2, \tag{34}$$

with

$$\Pi^{(C)}_\phi(\omega,\mathbf{k}) = \sigma_0^2\,\frac{\mathcal{N}_C(k^2)}{\mathcal{M}_C(k^2)\omega^2 + \mathcal{B}_C(k^2)}, \tag{35}$$

where

$$\mathcal{N}_C(k^2) \equiv \mathcal{M}_C(k^2)\mathcal{B}_C(k^2) - \Gamma_C(k^2)^2. \tag{36}$$

Equivalently,

$$\mathcal{N}_C(k^2) = m_S^2\mathcal{B}(k^2) + \frac{\kappa_\theta}{\mathcal{D}_C(k^2)}\Big(\alpha^2\mathcal{B}(k^2) + m_S^2\mathcal{Z}(k^2)^2\Big). \tag{37}$$

Proof. Because $\Gamma_C(-\nabla^2)$ depends only on the spatial operator, it commutes with ∂_0. Thus

$$\sigma\,\Gamma_C\,\partial_0\sigma = \frac{1}{2}\partial_0(\sigma\,\Gamma_C\,\sigma)\,,$$

which is a boundary term, giving (33). The remaining quadratic σ-action then has the same structure as in the earlier coefficient-level notes, but with the replacement

$$m_S^2 \mapsto \mathcal{M}_C, \qquad \mu_{SI} \mapsto \Gamma_C.$$

Integrating out σ therefore yields (35). Expanding $\mathcal{N}_C = \mathcal{M}_C\mathcal{B}_C - \Gamma_C^2$ with (29)–(31) gives (37). □

Remark 2. *The exact-closure burden is now carried by the generalized numerator $\mathcal{N}_C(k^2)$, not by $\mathcal{B}(k^2)$ alone. This is the precise sense in which the constraint-assisted branch changes the Phase 3 problem.*

To extract the low-momentum conditions, expand

$$\mathcal{G}(k^2) = G_0 - G_2 k^2 + \mathcal{O}(k^4), \tag{38}$$
$$\mathcal{Z}(k^2) = Z_0 - Z_2 k^2 + \mathcal{O}(k^4), \tag{39}$$
$$\mathcal{B}(k^2) = \mathcal{B}_0 + \mathcal{B}_2 k^2 + \mathcal{O}(k^4), \tag{40}$$

with

$$G_0 = \beta_I \widehat{M}^{-1\,IJ} \beta_J, \qquad G_2 = \beta_I \widehat{M}^{-2\,IJ} \beta_J, \tag{41}$$
$$Z_0 = \mu_{SI} \widehat{M}^{-1\,IJ} \beta_J, \qquad Z_2 = \mu_{SI} \widehat{M}^{-2\,IJ} \beta_J, \tag{42}$$
$$\mathcal{B}_0 = \mu_S^2 - \mu_{SI} \widehat{M}^{-1\,IJ} \mu_{SJ}, \tag{43}$$
$$\mathcal{B}_2 = \kappa_S + \mu_{SI} \widehat{M}^{-2\,IJ} \mu_{SJ}. \tag{44}$$

Define also

$$\mathcal{D}_{C,0} \equiv \Delta_C + \kappa_\theta G_0 = 1 - \kappa_\theta M_C^2 + \kappa_\theta \beta_I \widehat{M}^{-1\,IJ} \beta_J. \tag{45}$$

Proposition 6 (Infrared coefficients of the generalized numerator)**.** *On the branch with $\mathcal{D}_{C,0} \neq 0$, the generalized numerator expands as*

$$\mathcal{N}_C(k^2) = \mathcal{N}_{C,0} + \mathcal{N}_{C,2} k^2 + \mathcal{O}(k^4), \tag{46}$$

where

$$\mathcal{N}_{C,0} = m_S^2 \mathcal{B}_0 + \frac{\kappa_\theta}{\mathcal{D}_{C,0}} (\alpha^2 \mathcal{B}_0 + m_S^2 Z_0^2), \tag{47}$$
$$\mathcal{N}_{C,2} = m_S^2 \mathcal{B}_2 + \frac{\kappa_\theta}{\mathcal{D}_{C,0}} (\alpha^2 \mathcal{B}_2 - 2 m_S^2 Z_0 Z_2) + \frac{\kappa_\theta^2 G_2}{\mathcal{D}_{C,0}^2} (\alpha^2 \mathcal{B}_0 + m_S^2 Z_0^2). \tag{48}$$

Proof. Using (38) gives

$$\mathcal{D}_C(k^2) = \mathcal{D}_{C,0} - \kappa_\theta G_2 k^2 + \mathcal{O}(k^4),$$

and therefore

$$\frac{1}{\mathcal{D}_C(k^2)} = \frac{1}{\mathcal{D}_{C,0}} + \frac{\kappa_\theta G_2}{\mathcal{D}_{C,0}^2} k^2 + \mathcal{O}(k^4).$$

Substituting this together with (38)–(40) into (37) gives (47) and (48). □

Theorem 1 (Generalized flat-background exact-closure conditions)**.** *On the generic nonresonant constraint-assisted branch with $\mathcal{D}_{C,0} \neq 0$, quadratic infrared exact closure requires*

$$\mathcal{N}_{C,0} = 0, \qquad \mathcal{N}_{C,2} = 0. \tag{49}$$

Equivalently, the generalized coefficient conditions are

$$(m_S^2 \mathcal{D}_{C,0} + \alpha^2 \kappa_\theta) \mathcal{B}_0 + m_S^2 \kappa_\theta Z_0^2 = 0, \tag{50}$$

and

$$(m_S^2 \mathcal{D}_{C,0} + \alpha^2 \kappa_\theta) \mathcal{B}_2 - 2 m_S^2 \kappa_\theta Z_0 Z_2 - m_S^2 \kappa_\theta G_2 \mathcal{B}_0 = 0. \tag{51}$$

Proof. If $\mathcal{N}_{C,0} \neq 0$, then (35) contains an unsuppressed p^0 scalar bridge correction. If $\mathcal{N}_{C,0} = 0$ but $\mathcal{N}_{C,2} \neq 0$, then an unsuppressed k^2 non-Einsteinian scalar correction remains. Therefore quadratic infrared exact closure requires (49). Multiplying (47)=0 by $\mathcal{D}_{C,0}$ gives (50). Then substituting (50) into (48)=0 yields (51). □

Remark 3. *Equations* (50)–(51) *complete the fixed task for the constraint-assisted branch: they are the generalized exact-closure conditions replacing the old* $\mathcal{B}_0 = 0$, $\mathcal{B}_2 = 0$ *pair. They do* not *yet decide whether a healthy nondegenerate solution exists. That requires the next-stage sign and pole analysis of* $\mathcal{D}_C$, $\mathcal{M}_C$, *and* $\mathcal{B}_C$.

7. Conclusion

The constraint-assisted longitudinal U-sector family has now been carried through its full flat-background quadratic elimination. The (χ, C) sector does not collapse to a vanishing bracket as it did in the previous branches. Instead it feeds back into the scalar bridge channel as a rank-one deformation of the Q-sector operator.

That change is now fully explicit. After elimination of the coupled auxiliary system, the scalar bridge self-energy is governed by the generalized numerator $\mathcal{N}_C(k^2)$, not by the old operator $\mathcal{B}(k^2)$ alone. On the generic nonresonant branch, the quadratic exact-closure conditions are the two infrared coefficient relations (50)–(51).

This completes the immediate derivational task that the branch-selection note left open. The next task is sharper and narrower: determine whether the new conditions admit any healthy nondegenerate branch or whether the sourced longitudinal U-sector again contracts onto a resonant or otherwise degenerate solution. Until that is settled, exact closure remains the benchmark and reversion theory of the project.